

WASP-1b

R-1594

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-1_150804 · created 2026-08-02T01:24:48+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457238.8564 +/- 0.007 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.139 +/- 0.04
Transit depth (Rp/Rs)^2	1.94 +/- 1.12 [%]
Orbital Inclination (inc)	86.27 +/- 1.44
Airmass coefficient 1 (a1)	0.25 +/- 0.16
Airmass coefficient 2 (a2)	0.91 +/- 0.32
Scatter in the residuals of the lightcurve fit is	73.22 %
Best Comparison Star	None
Optimal Aperture	0.99
Optimal Annulus	4.75
Transit Duration (day)	0.15 +/- 0.01

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.139 ± 0.04 vs 0.1036 (deviation 0.88σ)
Duration fit vs published	0.15 d vs 0.1567 d (deviation 0.66σ)
Mid-time O–C	17.4 min
Depth SNR	3.5
Residual scatter	73.22 %

Light curve

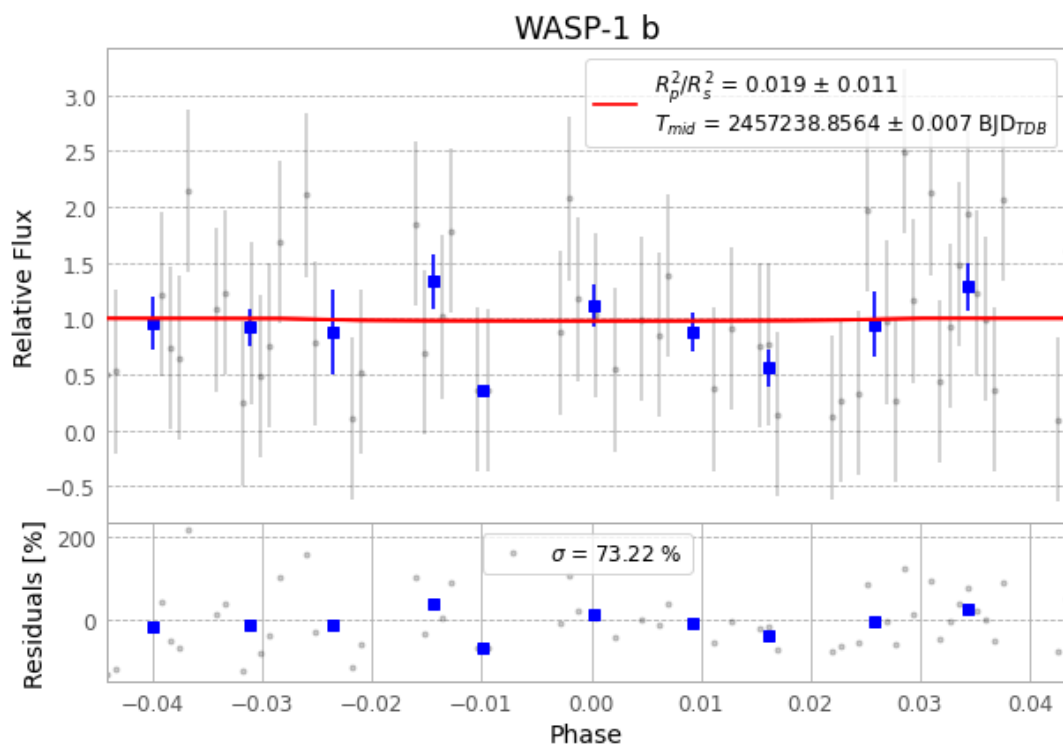


Figure 1 — FinalLightCurve_WASP-1 b_2015-08-03.png (SHA-256 f3fb184bc59651c1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [471, 260], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 1bc4eed2a11d6a7c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

107 FITS frames (70 MB), WASP-1150804054812.FITS → WASP-1150804110612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-1-150804.log (fda1b340e11f...), FinalLightCurve_WASP-1 b_2015-08-03.png (f3fb184bc596...), FinalLightCurve_WASP-1 b_2015-08-03.csv (0b51d754274d...)

Compute resources

Wall time 155.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 01:24Z.